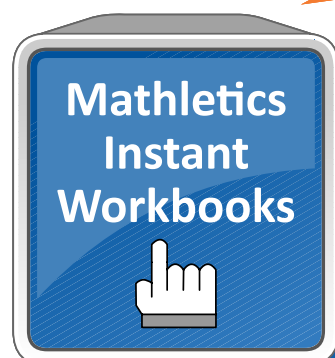


MATHLETICS

Percentages

Teacher Book - Series I-1

3.4% of 2560 = ____



Percentages

Topic Test

PART A

Instructions

This part consists of 12 multiple-choice questions

Each question is worth 1 mark

Fill in only ONE CIRCLE for each question

Calculators are NOT allowed

Time allowed: 15 minutes

Total marks = 12

						Marks
1	16 marks out of 64 is what as a percentage?					
	(A) 20%	(B) 24%	(C) 25%	(D) 30%	1	
2	An item is bought for \$80 and sold for \$120. Find the percentage profit on the cost price.					
	(A) 40%	(B) 30%	(C) 60%	(D) 50%	1	
3	18% of a number is 27. The number is					
	(A) 150	(B) 200	(C) 250	(D) 300	1	
4	As a decimal $4\frac{1}{2}\%$, equals					
	(A) 0.42	(B) 0.45	(C) 0.042	(D) 0.045	1	
5	Ted pays \$60 after receiving a discount of 20%. The normal price equals					
	(A) \$63	(B) \$72	(C) \$75	(D) \$80	1	
6	0.3% equals					
	(A) $\frac{1}{3}$	(B) $\frac{3}{10}$	(C) $\frac{3}{100}$	(D) $\frac{3}{1000}$	1	
7	At a roll call, there were 4 students absent and 21 present. What percentage of the roll class was absent?					
	(A) 16%	(B) 19%	(C) 25%	(D) 84%	1	
8	A car was bought for \$6000 and sold at a profit of 30%. What was its selling price?					
	(A) \$1800	(B) \$4200	(C) \$6200	(D) \$7800	1	
9	The price of a tyre is reduced from \$80 to \$72. What is the percentage discount?					
	(A) 8%	(B) 10%	(C) 11%	(D) 90%	1	
10	What percentage is \$3 of \$20?					
	(A) 10%	(B) 12%	(C) 15%	(D) 18%	1	
11	If 120 is 20% of a number, then the number is					
	(A) 24	(B) 96	(C) 480	(D) 600	1	
12	$p\%$ of \$ q is					
	(A) \$ pq	(B) \$ $(\frac{p+q}{100})$	(C) \$ $100pq$	(D) \$ $(\frac{pq}{100})$	1	

Total marks achieved for PART A

12

Percentages

Topic Test

PART B

Instructions This part consists of 15 questions
Each question is worth 1 mark
Write answers in the answers-only column

Time allowed: 20 minutes

Total marks = 15

Questions	Answers only	Marks
1 Find 37% of \$300.		1
2 Increase 260 kg by 10% and then decrease the result by 10%.		1
3 Write $\frac{7}{12}$ as a percentage correct to two decimal places.		1
4 If 15% of an amount is \$140, what is the total amount?		1
5 After 25% of a bill has been paid, \$80 still remains to be paid. How much was the bill?		1
6 A concrete mixture contains one part cement, two parts sand and three parts gravel. What percentage of the mixture is sand?		1
7 Express 700% as a fraction.		1
8 Find the simple interest on \$800 at 9% p.a. for 1 year.		1
9 Express 52 as a percentage of 416.		1
10 Find 100%, given that 40% is 98.		1
11 David pays 28% of his income in tax. If he pays \$14 200 income tax, what is his income?		1
12 A pen is bought for \$69.50 and sold for \$89.50. What is the gain per cent?		1
13 Find 70% of \$8640 and decrease this amount by 15%.		1
14 \$4800 is invested in a fund giving 12% interest per year. How much interest is received for 5 years?		1
15 The price of a television set is reduced by 18% to \$1722. What was the original price?		1

Total marks achieved for PART B

15

Answers – Percentages

PAGE 1 1 a $\frac{1}{2}$ b $\frac{2}{5}$ c $\frac{3}{10}$ d $\frac{1}{10}$ e $\frac{1}{5}$ f $\frac{3}{5}$ g $\frac{7}{10}$ h $\frac{4}{5}$ i $\frac{1}{4}$ j $\frac{3}{4}$ k $\frac{17}{20}$ l 1 m $\frac{17}{100}$ n $\frac{9}{10}$ o $\frac{19}{20}$ p $\frac{49}{100}$ q $\frac{19}{100}$ r $\frac{23}{100}$ 2 a $\frac{1}{40}$ b $\frac{41}{500}$ c $\frac{9}{200}$
d $\frac{73}{200}$ e $\frac{7}{200}$ f $\frac{11}{40}$ g $\frac{169}{200}$ h $\frac{201}{500}$ i $\frac{5}{8}$ j $\frac{1}{200}$ k $\frac{9}{40}$ l $\frac{1}{125}$ m $\frac{21}{200}$ n $\frac{17}{200}$ o $\frac{183}{1000}$ p $\frac{37}{400}$ q $\frac{81}{200}$ r $\frac{131}{200}$ 3 a $\frac{1}{200}$ b $\frac{1}{300}$ c $\frac{1}{400}$ d $\frac{1}{800}$ e $\frac{9}{40}$ f $\frac{21}{40}$
g $\frac{11}{200}$ h $\frac{1}{3}$ i $\frac{2}{3}$ j $\frac{17}{200}$ k $\frac{11}{200}$ l $\frac{1}{8}$ m $\frac{21}{200}$ n $\frac{51}{200}$ o $\frac{71}{200}$ p $\frac{171}{200}$ q $\frac{31}{200}$ r $\frac{81}{400}$ 4 a $\frac{19}{50}$ b $\frac{14}{25}$ c $\frac{31}{50}$ d $\frac{19}{75}$ e $\frac{49}{400}$ f $\frac{13}{400}$ g $\frac{19}{200}$ h $\frac{43}{500}$ i $\frac{6}{125}$
j $\frac{37}{1000}$ k $\frac{69}{1000}$ l $\frac{13}{250}$ m $\frac{21}{50}$ n $\frac{11}{20}$ o $\frac{18}{25}$

PAGE 2 1 a 30% b 50% c 80% d 40% e 43% f 84% g 62% h 39% i 56% j 25% k 16% l 72% 2 a 50% b 160% c 70%
d 80% e 30% f 18% g 36% h 28% i 14% 3 a 75% b $6\frac{1}{4}\%$ c 65% d $11\frac{2}{3}\%$ e 25% f $17\frac{1}{2}\%$ g 15% h 60% i $17\frac{1}{2}\%$
4 a $37\frac{1}{2}\%$ b $55\frac{5}{9}\%$ c $62\frac{1}{2}\%$ d $63\frac{7}{11}\%$ e $41\frac{2}{3}\%$ f $33\frac{1}{3}\%$

PAGE 3 1 a 0.1 b 0.8 c 0.3 d 0.5 e 0.4 f 0.6 g 0.2 h 0.7 i 0.9 j 1 k 0.15 l 0.65 2 a 0.16 b 0.19 c 0.39 d 0.12 e 0.61
f 0.56 g 0.14 h 0.73 i 0.67 j 0.98 k 0.81 l 0.78 m 0.62 n 0.43 o 0.49 3 a 0.05 b 0.02 c 0.08 d 0.09 e 0.06 f 0.07 g 0.03
h 0.01 i 0.04 j 0.015 k 0.025 l 0.033 m 0.021 n 0.017 o 0.049 4 a 0.455 b 0.892 c 0.574 d 0.639 e 0.632 f 0.365 g 0.538
h 0.927 i 0.697 j 0.237 k 0.158 l 0.213 5 a 0.3 b 0.0083 c 0.00375 d 0.015 e 0.6 f 0.125 g 0.0925 h 0.2075 i 0.007

PAGE 4 1 a 15% b 25% c 35% d 95% e 14% f 85% g 18% h 55% i 17% j 45% k 67% l 65% m 91% n 75% o 72%
2 a 80% b 60% c 10% d 50% e 90% f 20% g 70% h 40% i 30% j 1% k 2% l 5% m 4% n 6% o 3% 3 a 12.3% b 62.5%
c 53.7% d 18.9% e 42.7% f 13.9% g 83.5% h 56.7% i 68.3% j 12.9% k 81.6% l 92.7% m 68.7% n 25.7% o 36.9%
4 a 0.1% b 91% c 0.6% d 0.7% e 0.2% f 0.9% g 81% h 3.7% i 2.1% j 0.4% k 0.5% l 0.3% m 5.7% n 8.9% o 6.5%
5 a 120% b 250% c 360% d 256% e 131% f 357% g 825% h 341% i 563% j 284% k 534% l 813% m 527% n 803% o 206%

PAGE 5 1 a 30 b 1000 c 270 d 252 e 290 f 630 g 92.5 h 50.4 i 10.5 j 135 k 623 l 79 m 56.25 n 664 o 313.65 p 650
2 a 0.48 b 8.84 c 0.9 d 7.82 e 5.727 f 207.936 g 2.365 h 677.875 i 556.25 j 302.95 k 11 l 105 m 45 n 516.6 o 60 p 46.8
3 a 85 b \$224 c 6.03 d 67.71 days e 46.2 f \$2 g \$18 h 1.2kg i 80c j 120 k 200g l \$203 m \$60 n \$170.40 o 45 p 68

PAGE 6 1 a \$58.50 b \$9200 c \$116 d \$4800 e \$420 f \$1012.50 g 1260g h 1 hour i 640kg j 360L k 72h l 140t m \$810
n \$372.13 o \$209.95 p \$168.64 q 96ha r 1020 cm 2 a \$51.03 b \$57.12 c 2.6m d 197.6L e \$837 f \$5083.20 g \$100.50 h 402.50
i \$789.23 j \$121.90 k 36h l \$252 m \$310.25 n \$528 o \$57.60 p \$12.10 q 676.80 r \$403 3 a \$414 000 b \$1800 c \$24000 d \$97.50

PAGE 7 1 a 10% b 40% c $57\frac{1}{7}\%$ d $35\frac{5}{7}\%$ e $36\frac{4}{11}\%$ 2 a 25% b 25% c 40% d 25% e 50% f $33\frac{1}{3}\%$ g 12.5%
h 90% i 25% j 5% k $8\frac{1}{3}\%$ l 63% m $66\frac{2}{3}\%$ n 7.5% 3 a 12.5% b $16\frac{2}{3}\%$ c $13\frac{1}{3}\%$ d 2% e 20% f 25% g $33\frac{1}{3}\%$ h 25%
i 30% j 10% k 25% l 40% m 4% n 1.6% o $83\frac{1}{3}\%$ p $6\frac{2}{3}\%$ 4 a $8\frac{1}{3}\%$ b 25% c $15\frac{5}{9}\%$ d $12\frac{1}{2}\%$ e $16\frac{1}{4}\%$ f $87\frac{1}{2}\%$
g 20% h $333\frac{1}{3}\%$ i 2% j 25% k $55\frac{5}{9}\%$ l 30% m $7\frac{1}{2}\%$ n 25%

PAGE 8 1 a 64 b \$150 c 50 d 40 2 a 160 b 630 c 2400 d 700 3 a 200 b 460 c 480 d 400 e 140 f 600 g 1660 h 264.4
i 4700 j 425 k 287.5 l 1840 4 a 27 b \$21500 c \$575 d \$168000 e \$550 f \$54.17 g 5000

PAGE 9 1 \$360 2 504 3 78 4 75% 5 \$99 6 200 7 94.69% 8 \$2950 9 \$190 10 \$881.10 11 \$969.50 12 3 13 a 51.46%
b 33.97% 14 50%, 40%, 10%, 845 blue tiles 15 \$253.83

PAGE 10 1 C 2 D 3 A 4 D 5 C 6 D 7 A 8 D 9 B 10 C 11 D 12 D

PAGE 11 1 \$111 2 257.40kg 3 58.33% 4 \$933.33 5 \$106.67 6 $33\frac{1}{3}\%$ 7 7 8 \$72 9 12.5% 10 245 11 \$50 714.29 12 28.78%
13 \$5140.80 14 \$2880 15 \$2100